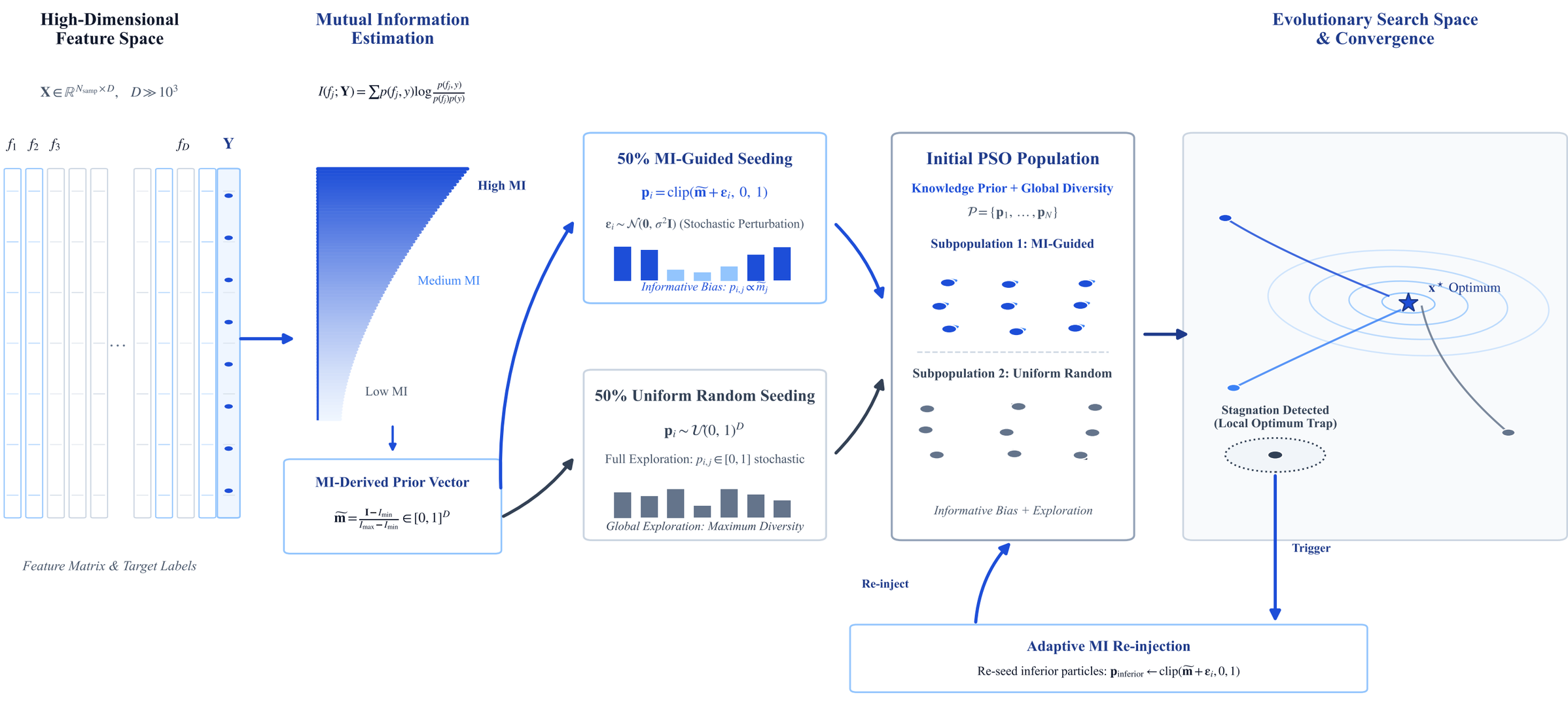

High-Dimensional Feature Space
$\mathbf{X} \in \mathbb{R}^{N_{\text{samp}} \times D}, \quad D \gg 10^3$
f_1 f_2 f_3 ... f_D Y
Feature Matrix & Target Labels
Mutual Information Estimation
$I(f_j; \mathbf{Y}) = \sum p(f_j, y) \log \frac{p(f_j, y)}{p(f_j)p(y)}$
High MI
Medium MI
Low MI
MI-Derived Prior Vector
$\widetilde{\mathbf{m}} = \frac{\mathbf{I} - I_{\min}}{I_{\max} - I_{\min}} \in [0, 1]^D$
50% MI-Guided Seeding
$\mathbf{p}_i = \text{clip}(\widetilde{\mathbf{m}} + \boldsymbol{\varepsilon}_i, 0, 1)$
$\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ (Stochastic Perturbation)
Informative Bias: $p_{i,j} \propto \widetilde{m}_j$
50% Uniform Random Seeding
$\mathbf{p}_i \sim \mathcal{U}(0, 1)^D$
Full Exploration: $p_{i,j} \in [0, 1]$ stochastic
Global Exploration: Maximum Diversity
Initial PSO Population
Knowledge Prior + Global Diversity
$\mathcal{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_N\}$
Subpopulation 1: MI-Guided
Subpopulation 2: Uniform Random
Informative Bias + Exploration
Evolutionary Search Space & Convergence
$\mathbf{x}^\star$ Optimum
Stagnation Detected (Local Optimum Trap)
Trigger
Re-inject
Adaptive MI Re-injection
Re-seed inferior particles: $\mathbf{p}_{\text{inferior}} \leftarrow \text{clip}(\widetilde{\mathbf{m}} + \boldsymbol{\varepsilon}_i, 0, 1)$
Raw High-D Data
MI-Derived Prior
Hybrid Population Seeding
Knowledge-Biased Swarm
Stagnation Resilience